Real-time person identification — on a laptop, offline, since 2023.

Narrow AI — the model solves one task (detecting people in the frame) and nothing else.

A model frame during the demo: 2 people in boxes. YOLOv8 (Ultralytics, 2023). [1]

On screen — **YOLOv8** on a laptop CPU: **~30 fps**.

Offline · trained in 2023. [2]

[1] *Ultralytics — YOLOv8 (2023)* · **[2]** *Google — MediaPipe*

Welcome to the course

«Industry Applications of Artificial Intelligence Systems»

17 lectures on where AI works across industries — and where it does not.

The course's core question is not «can we use AI?» but «should we, and where?».

100% of AI pilots launch

−90% get rolled back

10% reach production

An illustration of the principle (Gartner, McKinsey confirm similar numbers). [1]

The core idea of the course

Tomorrow — almost everywhere.
Today — almost no one.

The course is about that gap. [2]

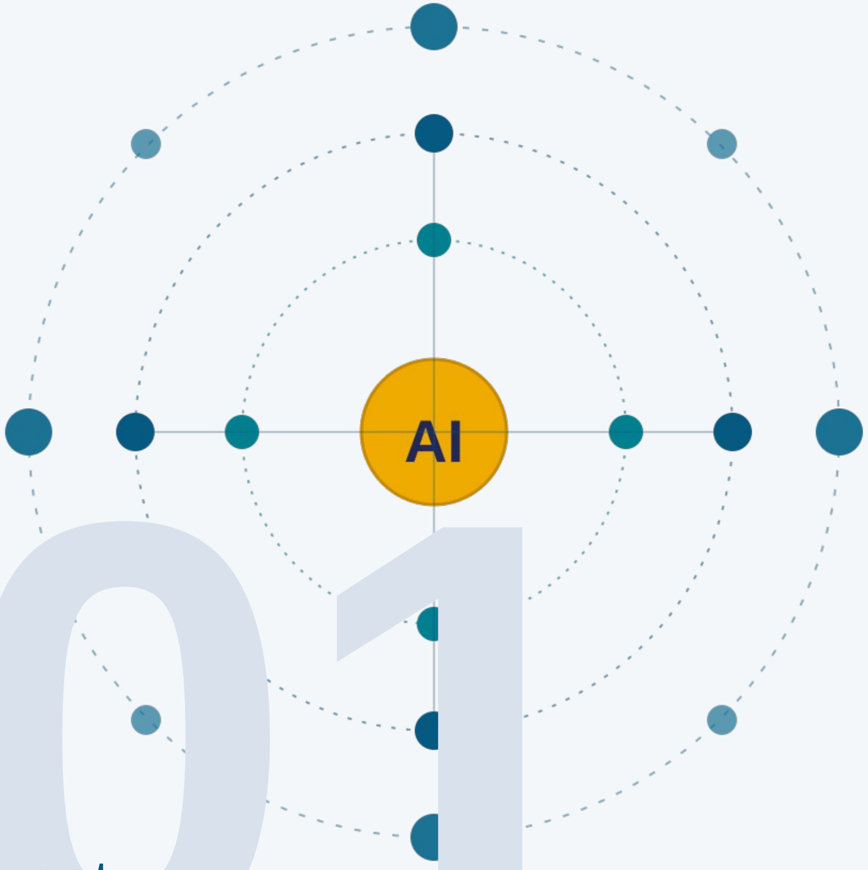
The central question of the course

Where does AI work,
where does it not,
and how do you tell?

[1] Gartner - press release (Oct 2024) · [2] Vedomosti / Intellectual Analytics (Mar 2026)

LECTURE

What is AI?
History, classification, core concepts



01

A map of AI applications: where it works, where it does not.

Lecture plan

0	1	2	3	4	5
Intro	**What is AI**	**Where we are now**	**4 ways to implement**	**Limits and safety**	**Wrap-up**
Demo · instructor · central question	*Definitions, history, turning point*	*Market numbers 2022–2026*	*Model · chat · agent · app*	*What AI breaks and where it fails*	*Recap · semester map*

Maxim Levko

Telegram
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Email
Levko.maxim@gmail.com

Who I am and why this matters to me.

Architect, technical and product lead for building and deploying information systems

20+ years of experience in IT

10+ completed projects under my lead

Expertise

Systems analysis · Systems design · Data management · Business automation · Product management

Consulting and in-house

- Yandex
- MTS
- Magnit
- Sibur

Section 1 · What is AI

Definitions, history, classification.

0	**1**	**2**	**3**	**4**	**5**
Intro	**What is AI**	**Where we are now**	**Four ways to implement**	**Limits and safety**	**Wrap-up**
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There are many definitions of AI — because AI is a moving target.

Russell & Norvig (AIMA, 2021)

«AI = a system that thinks like a human, thinks rationally, acts like a human, or acts rationally (4 quadrants on 2 axes).»

Russell & Norvig, AIMA, 4th ed., 2021 [1]

ISO/IEC 22989:2022

«An AI system is an engineered system that generates outputs (recommendations, predictions, decisions) for human-defined goals.»

International standard ISO/IEC 22989:2022 — basis of the EU AI Act [2]

Via learning (Mitchell, 1997)

«A program improves with experience E on task T by measure P. If behavior emerges from a trained model — that is AI.»

Mitchell, Machine Learning, 1997 [3]

Via benchmarks and AGI

«AI = whatever passes the Turing test or solves a benchmark at human level.» Searle's objection: behavior != understanding.

Turing 1950 / Searle 1980 — Chinese Room [4]

AI Effect (Tesler): «AI is whatever hasn't been done yet». Once a technique starts working, people stop calling it AI.

[1] Russell & Norvig — AIMA, 4th ed. (2021) · **[2]** ISO/IEC 22989:2022 · **[3]** Mitchell — Machine Learning (1997) · **[4]** Searle — Chinese Room (1980) / Turing (1950)

The idea of a neural network is 13 years older than the term «artificial intelligence» itself.

1943

McCulloch and Pitts [1]:
the formal neuron
as a logic element

13 years

1956

Dartmouth conference:
the term «artificial intelligence»

Pitts (right) and Lettvin, 1959 — the frog experiment at MIT (this is NOT a 1943 photo). Wikimedia Commons, CC BY-SA 3.0.

The formal neuron solved no applied tasks, but it anticipated the connectionist tradition — the line of thought from which neural networks and the transformer would grow decades later.

The idea of a «neural network», at the theoretical level, is older than the term «artificial intelligence» itself.

[1] McCulloch & Pitts (1943)

70 years of AI: breakthroughs, winters, the 2017 turning point. [2] [3]

Breakthroughs

- Turing — a test for thinking — **1950**
- ELIZA — Weizenbaum — **1966**
- Expert systems — **1980s**

Winters and leaps

funding leaves when promises fall through

- 1st winter — the Lighthill report — **1974**
- Deep Blue — 200M pos/sec — **1997**
- AlexNet — GPU + DL — **2012**

Turning point and boom

- **«Attention Is All You Need» ★** — **2017**
- ChatGPT — 1M in 5 days
- DeepSeek R1, Claude Code

★ 2017 — Vaswani and 7 co-authors (Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, Polosukhin) introduce the self-attention mechanism — the basis of all modern LLMs. As of May 2026 the paper has over 160,000 citations. [1]

[1] Vaswani et al. — Attention Is All You Need (2017) · [2] McCorduck — Machines Who Think (2004) · [3] Dhar — Paradigm Shifts in AI, CACM (2024)

Section 2 · Where we are now

AI as infrastructure, breakthroughs 2023–2026.

0	1	2	3	4	5
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AI became infrastructure in 3 years: 900M users, 51% of developers daily, 46% of Copilot code.

900M

WAU

ChatGPT, February 2026 [2]

OpenAI

51%

daily

Stack Overflow Dev Survey 2025 [1]

n=49k+, 177 countries

46%

of Copilot code

GitHub Octoverse 2025 [3]

Java — 61%

$390.9B → $539.5B

AI market 2025 → 2026

Grand View Research, 2026 [4]

Statista (software-only): ~$244–260B

Counter-fact: ~90% of AI pilots in Russia never reach production. CNews / Vedomosti / Intellectual Analytics, March 2026. [5]

[1] Stack Overflow — Developer Survey 2025 · [2] OpenAI — ChatGPT WAU (Feb 2026) · [3] GitHub — Octoverse 2025 · [4] Grand View Research (2026) · [5] Vedomosti / Intellectual Analytics (Mar 2026) · [6] Gartner - press release (Oct 2024)

The space is open: 4 breakthroughs of 2023–2026 from non-incumbent players.

September 2023	January 2025	November 2025	February 2026
Mistral 7B	**DeepSeek R1**	**OpenClaw**	**llama.cpp**
Apache 2.0 beats Llama-2 13B	**$589B Nvidia drop in a day**	100K★ stars in a quarter	100K+★ on GitHub faster than PyTorch
Mistral AI (FR) [1]	*DeepSeek (CN)* [2, 3, 4]	*P. Steinberger* [5]	*G. Gerganov / ggml.ai* [6]

Don't despair: serious breakthroughs come from different teams. This course is about durable concepts that survive model-generation turnover.

[1] Mistral AI — Announcing Mistral 7B (Sep 2023) · [2] DeepSeek-R1 — tech report (Jan 2025) · [3] SemiAnalysis — DeepSeek cost analysis (2025) · [4] Bloomberg — Nvidia $589B drop (27 Jan 2025) · [5] Steinberger — OpenClaw (GitHub) · [6] Gerganov — llama.cpp / ggml.ai

Section 3 · Four ways to implement AI systems

Not alternatives, but layers.

0	1	2	3	4	5
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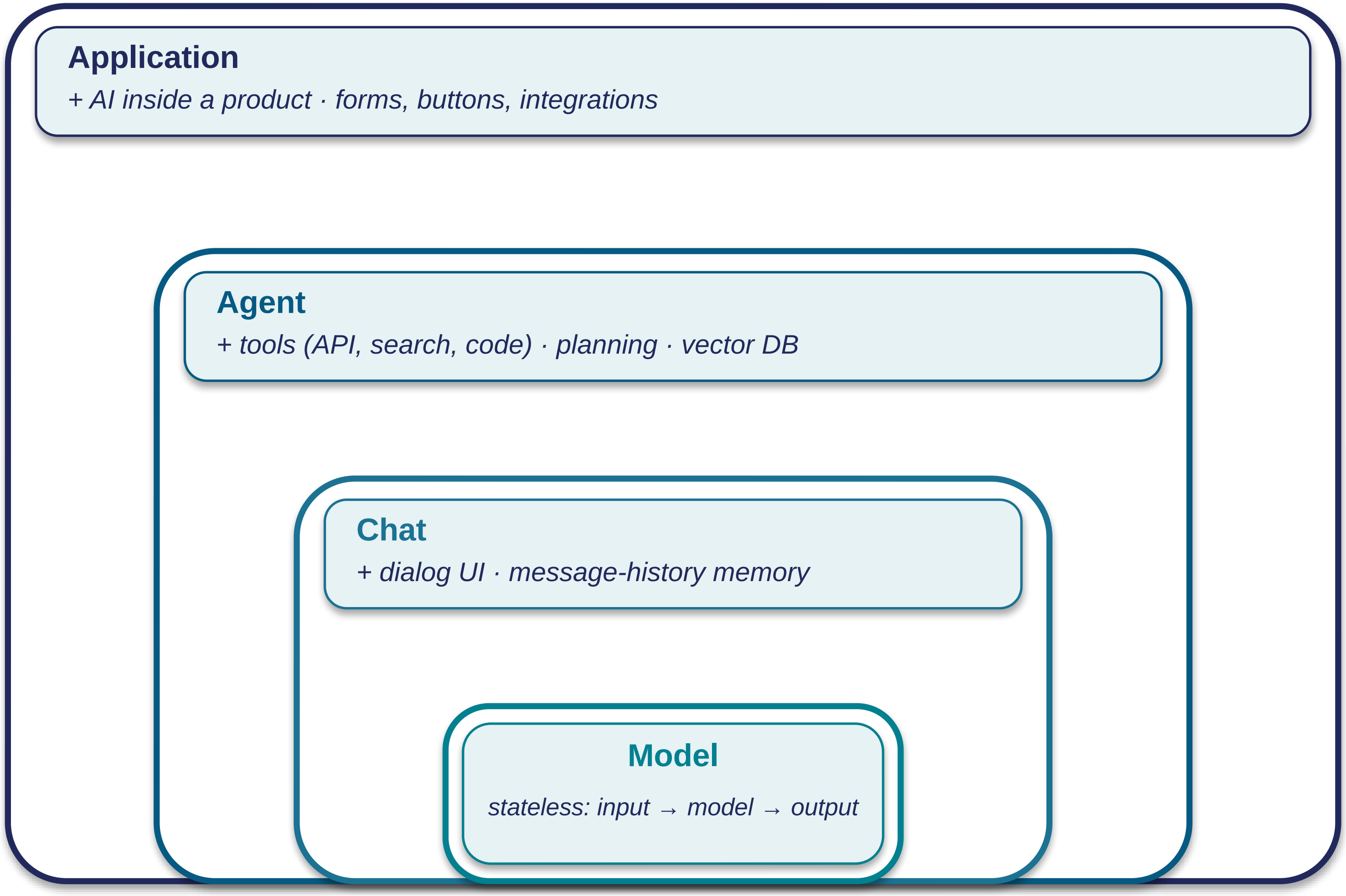
Ways to implement AI systems: not alternatives, but layers.

Each next layer

contains the previous one

Not four alternative technologies, but four ways to implement one task. [1] Each additional wrapper adds capability — and complexity, cost, and room for error. [2]



Choosing a layer is an engineering decision, not an either/or.

[1] Anthropic — Building Effective Agents (2024) · [2] Weng — LLM Powered Autonomous Agents (2023)

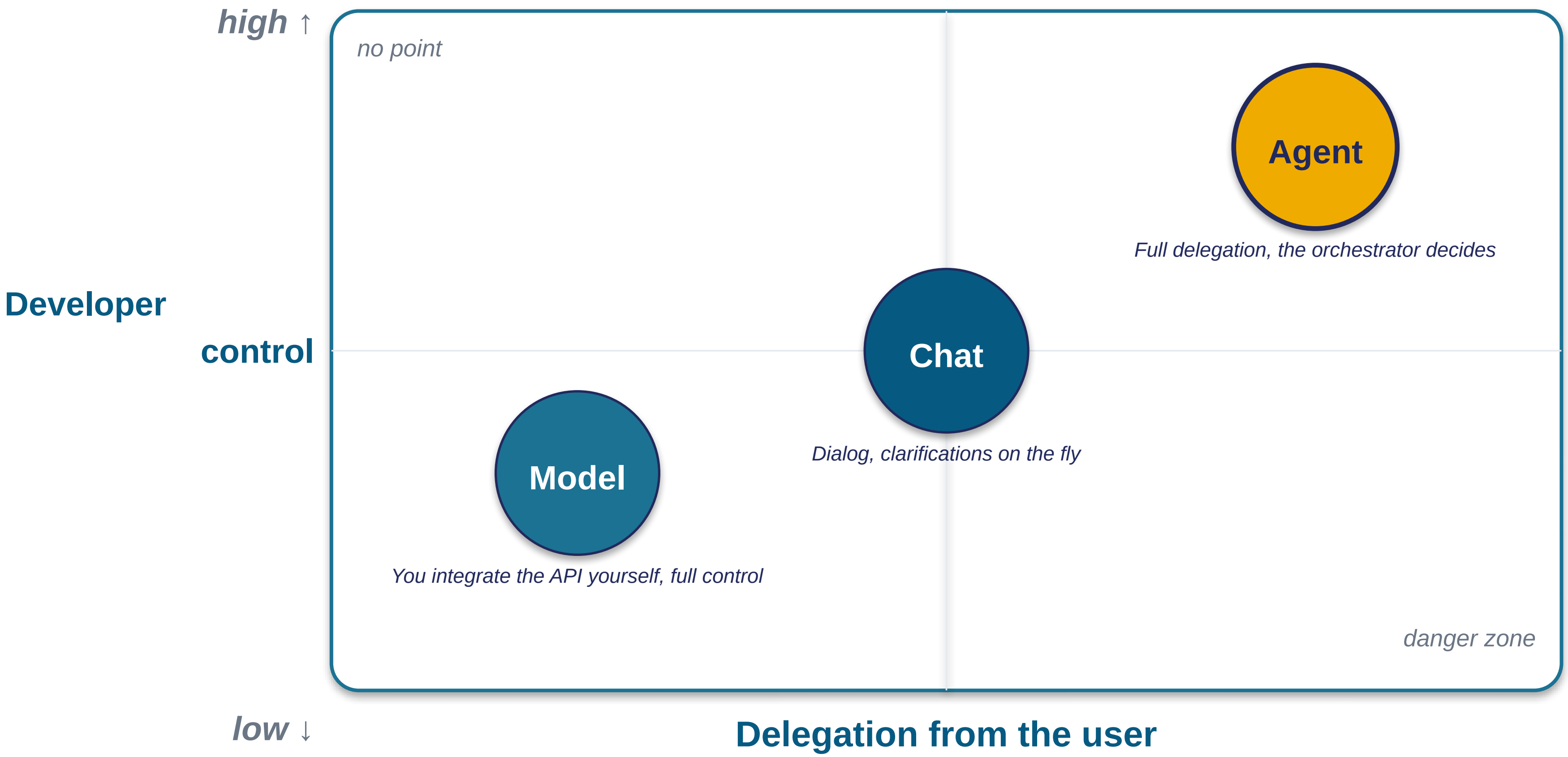
Classifying AI systems [1] — two axes: task type × modality. [2]

	Classif.	Recogn.	Retrieval	Generat.	Forecast
Text	BERT, spam	spaCy NER	BM25	GPT-4o, Claude	GPT-4o, Claude
Image	ResNet	YOLO	CLIP	DALL-E, MJ	frame forecast
Audio / video	PANNs	Whisper	Shazam	ElevenLabs, Sora	video forecast
Struct. data	XGBoost	table OCR	vector DB	Codex tab.	Prophet, ARIMA

YOLO — the object detection from the demo at the start of the lecture. Its training approach and architecture come later in the course.

[1] Russell & Norvig — AIMA (2021) · [2] Goodfellow, Bengio, Courville — Deep Learning (2016)

One task, three ways: control is split between the developer and the user.



The same task

Extract fields from an incoming PDF contract:

- signing date
- counterparty
- amount
- term

and put them into a table.

The distribution of control is an engineering decision, not one way being superior to another. [1]

[1] Anthropic — Building Effective Agents (2024)

MODEL

The model is a component, not a system. Inference: input → preprocessing → model → postprocessing → output.

This is already an application

Raw input
camera frame, text, audio
↑ *external system*

→

Prepro-cessing
scaling, cropping, tokenization
↑ ***developer***

→

Model
inference
↑ ***AI model***

→

Postpro-cessing
filtering, normalization
↑ ***developer***

→

Output
JSON, label, action
↑ *application*

YOLOv8
object detection on images

Whisper
speech recognition

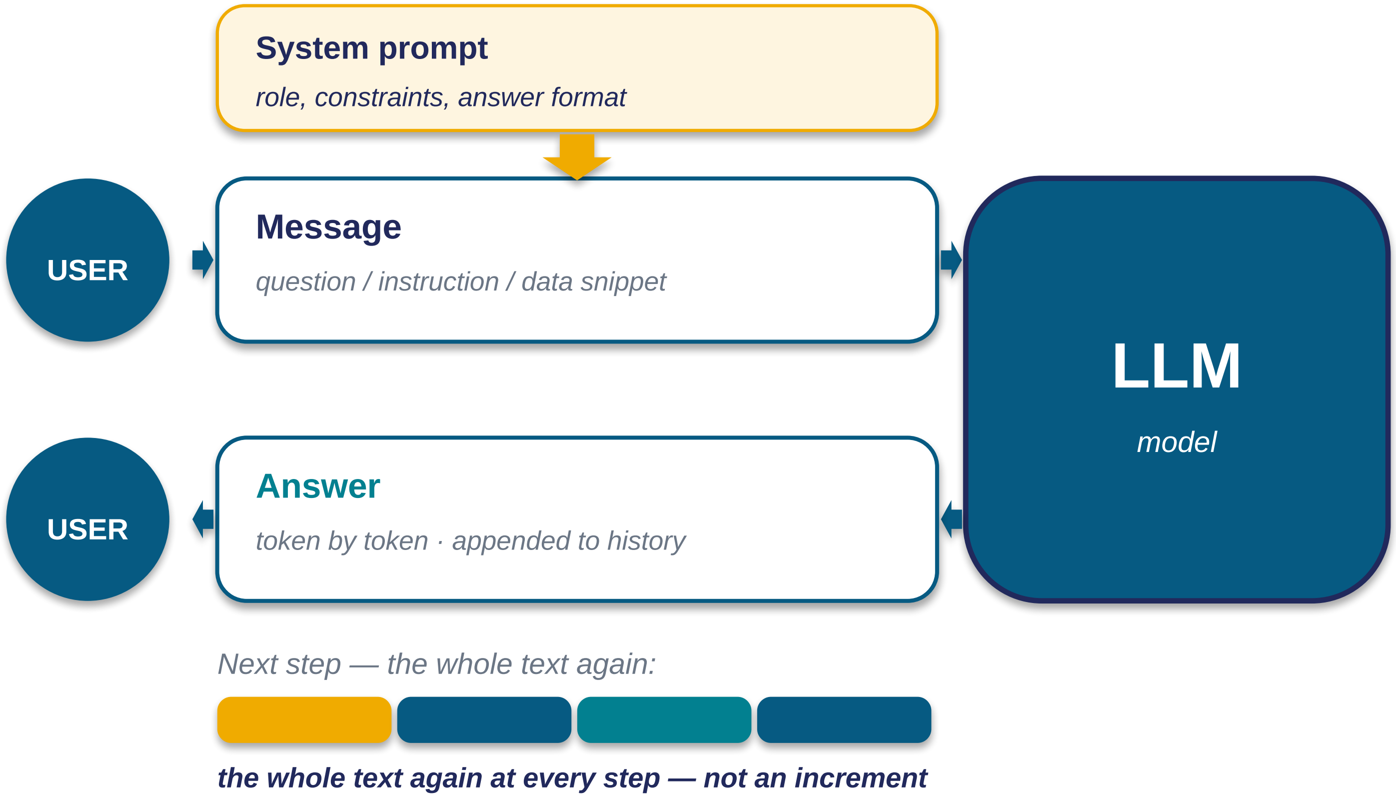
Stable Diffusion
image generation

AlphaFold
protein-structure prediction [2]

Preprocessing and postprocessing are the developer's responsibility. [1] YOLO = 50 lines; a working system with YOLO = hundreds of lines.

[1] Kreuzberger et al. — MLOps (2023) · [2] Jumper et al. — AlphaFold, Nature (2021)

How chat works: the dialog cycle.



Control via the system prompt.
The prompt sets role, constraints, format. The same chat is tuned for different scenarios — that is an engineering lever for the developer. [1]

The limit is the context window.
128k–1M tokens; when the history no longer fits, old messages drop out and the chat «forgets» the start of a long conversation. This is a model limitation, not a bug.

Chat is a pipeline: «assemble → feed → append → show».

[1] Anthropic — Building Effective Agents (2024)

CHAT

Chat = model + interface + dialog memory.

A case — typical for chat

An engineer received an unclear spec from an adjacent team and needs to build a checklist for their own work.

You: Explain clause 4.2 of the spec in plain words.

Chat: This is a requirement for…

You: Build a 5-item checklist.

Chat: 1. Check… 2. Confirm… 3. …

A one-off task with clarifications — optimal for chat. Not a model, not an agent, not an

A caveat for production systems

Pure chats are almost never used in production.

Almost everywhere they are extended into agents [1] [2] — at least for long-term memory and search over a corporate knowledge base (RAG).

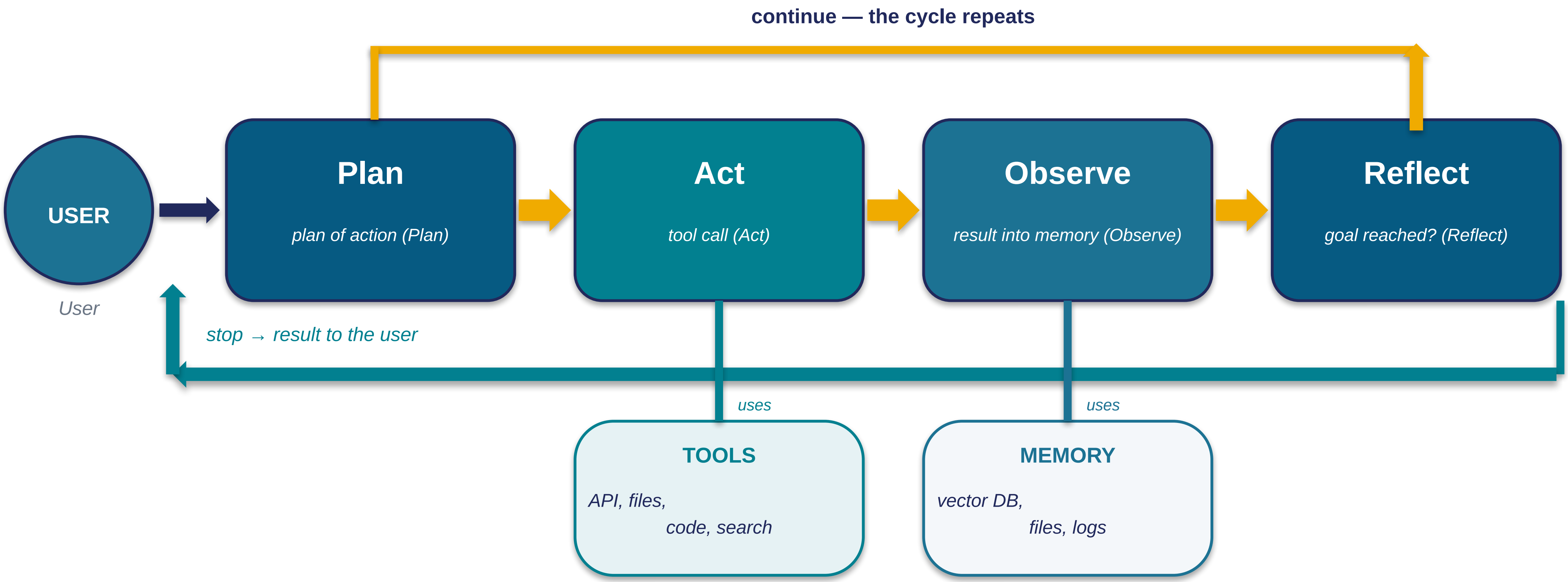
We'll break down the agent architecture on the next slide.

Choosing chat is a point on the interaction scale, not the one right answer.

[1] VCIOM — «Neural networks in Russians' lives» (Oct 2025) · [2] Dam et al. — Survey on LLM-based AI Chatbots (2024)

AGENT

Agent = chat + orchestrator + external memory + tools. [1] [2] [3]



ReAct (Reasoning + Acting) — Yao et al. 2022 (arXiv:2210.03629)

[1] Weng — LLM Powered Autonomous Agents (2023) · **[2]** Anthropic — Building Effective Agents (2024) · **[3]** Anthropic — Model Context Protocol (2024)

AGENT

An agent at work: 200 PDFs — a sequence of steps.

A case — typical for an agent

200 PDF reports.
Extract date, counterparty, amount.

Build a summary table.

Not a model — there is no specialized model for «take 200 arbitrary PDFs».

Not a chat — copying 200 files into a window is painful.

An agent is the natural choice.

What the agent does — step by step, with the tool named

	Step	Tool
1	Get the file list	*file system*
2	Open PDF #1	*PDF reader*
3	Extract text	*text extraction (OCR / parser)*
4	Summarize → vector DB	*embeddings + vector database*
5	Find the key fields	*retrieval + LLM extraction*
6	Write a row into the table	*table write (Sheets API / CSV)*
7	Loop over all 200 files	*orchestrator loop*

An agent = a sequence of tool calls, orchestrated by an LLM. [1]

[1] Yao et al. — ReAct (2022)

Levels of autonomy for AI agents: a design decision, not a property of the model.

5 levels of autonomy (Feng / McDonald / Zhang, 2025) [1]

5. Observer
User: only receives the result
AutoGPT overnight

4. Approver
User: approves at checkpoints
agent opens a PR, waits for review

3. Consultant
User: sets the goal, edits the plan
Devin fixes a bug from a ticket

2. Collaborator
User: works alongside the agent
Cursor pair programming

1. Operator
User: approves every action
Claude Code «approve each»

Where the human sits relative to the loop

Human-out-of-the-loop
The human only sees the result (≈ level 5).

Human-on-the-loop
The human watches, interrupts on deviations (≈ levels 3-4).

Human-in-the-loop
The human approves every step (≈ levels 1-2).

Override modes
Any level — with a manual override on alert.

The autonomy level is a product choice, not a property of the model. In the seminars we'll choose it deliberately.

[1] Feng, McDonald, Zhang — Levels of Autonomy (2025)

An application = AI packaged into a product interface.

Google Translate — scale in 2026

1+ billion users per month · **1 trillion** words translated / month

across Google Translate, Search, Lens, and Circle to Search (Google Blog, April 2026) [1]

Google Translate

neural translation

Notion AI

GPT-4/Claude in buttons

YandexGPT in Search

AI quick answer

Grammarly

NLP + LLM suggestions

Yandex Maps

ML routing

Adobe Firefly

diffusion in Photoshop

AI as a feature, not a product. Most students already use all six every day. [2]

[1] Google — Translate at 20 (2026) · [2] Anthropic — Building Effective Agents (2024)

Checklist «Which type of AI to choose»: 2 questions + a 2×2 quadrant. [1] [2] [3]

QUESTION 1

Do you need interaction with the user?

QUESTION 2

Do you need autonomous work with tools?

	NO	YES
YES	**CHAT** *corp chat for parsing a spec*	**AGENT** *200 PDFs → summary table*
NO	**MODEL** *pipeline detector*	**APPLICATION** *ETL with an AI classifier (automation)*

[1] Anthropic — Building Effective Agents (2024) · [2] Google — AI Agents Whitepaper (2024) · [3] Ng — Four Agentic Design Patterns (2024)

Section 4 · The limits of AI — your zone of responsibility

Where the data goes · AI errors · «can't do it» is yours too. [1] [2] [3]

0	**1**	**2**	**3**	**4**	**5**
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[1] NIST — AI RMF 1.0 (2023) · [2] NIST — Generative AI Profile 600-1 (2024) · [3] EU AI Act — Reg. (EU) 2024/1689

Consumer vs enterprise plans — where your data goes.

From the general zone of responsibility to the first concrete risk: data.

CONSUMER PLANS

data → training by default

- ChatGPT Free / Plus — training by default
- Anthropic Claude (since Sep 2025) — opt-in, 5-year retention
- Gemini Free — training + human review, 3 years
- YandexGPT Free — standard policy

ENTERPRISE PLANS / API [2]

data ≠ training

- ChatGPT Enterprise / Business — no training on your data
- OpenAI API (since March 2023) — no training on your data
- Anthropic for Business — zero data retention available
- Google Workspace / Vertex AI — no training on your data

Samsung 2023 — the canonical incident [1]

3 episodes in a month (March–April): source code, a meeting transcript, test sequences → ended up in OpenAI's dataset. Samsung banned external GenAI.

EU AI Act — fines [3]

up to €15M / 3% of turnover

up to €35M / 7% — for prohibited practices

[1] Bloomberg — Samsung bans ChatGPT (May 2023) · [2] OpenAI — Enterprise Privacy / data usage (2025) · [3] EU AI Act — Reg. (EU) 2024/1689

Hallucinations are an inherent property of AI. [1] [2]

Prompt

«Name three 2023-2024 papers on "seismic resistance of buried pipelines" with authors, journal, and DOI».

AI answer (3 fake references):

Petrov A., Smith J. (2023).
Seismic Resilience of Small-Diameter Pipelines.
DOI: 10.1016/j.engfailanal.2023.107214 ✗

Ivanov K. et al. (2024).
Underground Infrastructure Earthquake Response.
DOI: 10.1080/15732479.2024.2218450 ✗

Chen L., Brown R. (2023).
Microscale Pipe Vibration Analysis.
DOI: 10.1007/s11069-023-06122-1 ✗

Vectara HHEM (2025-26) [3]

hallucination rate

< 1%
summarization (Gemini 2.0 Flash)

10–15%
reasoning (multi-step)

Anti-pattern: «AI knows everything». Any AI answer to a factual question is a hypothesis to verify. [4]

[1] Huang et al. — Survey on Hallucination in LLMs (2023) · [2] Ji et al. — Survey of Hallucination in NLG (2023) · [3] Vectara — HHEM Leaderboard · [4] CybSafe / NCA — Oh Behave! (2024–25)

Bias, sycophancy, distribution shift — three manifestations of one nature.

Bias

The model repeats the dataset's skews.

A résumé screener trained on historical data discriminates — not by «deciding», but statistically.

Sycophancy

The model learns from feedback labeling to agree.

It agrees with the clearly wrong, over-praises — the user stops noticing the loss of critical judgment.

Distribution shift

Data from a period goes stale.

A model trained on 2023 code will, in 2026, suggest an outdated library with no visible failure.

GPT-4o: sycophancy — April 2025 [1]

Apr 25 — update released → **Apr 28** — rollback begins (Altman posts on social media that evening) → **Apr 29** — root-cause postmortem

The shared cause: the model reflects the data it was trained on. [2]

[1] OpenAI — Sycophancy in GPT-4o postmortem (2025) · [2] Pan et al. — Reward Misspecification (2022)

AGI forecasts — 4 speakers, 4 different incentives. [1] [2]

Speaker	Affiliation	AGI forecast	Material interest
Sam Altman	OpenAI	«We know how to build AGI; the start of superintelligence» (Jan 2026)	$100B+ rounds; IPO context
Dario Amodei	Anthropic	«AGI in 2–3 years; Nobel level within 2 years» (Davos 2026)	Competition with OpenAI; 2026 round
Demis Hassabis	Google DeepMind	«AGI by 2029–2030 (3–4 years); the window narrowed over 2026» (Axios/Google I/O, May 2026)	Community leader; more credibility from a cautious stance
Yann LeCun	AMI Labs (ex-Meta)	«LLMs won't lead to AGI; we need world models, JEPA»	$1B round (March 2026) for an alternative path

[1] Searle — Minds, Brains, and Programs (1980) · [2] Bostrom — Superintelligence (2014)

Section 5 · What to take home

Recap · semester map · Lecture 2 teaser.

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What we covered: three key takeaways.

1 **AI is a spectrum, not a monolith**

Task type × modality × implementation type.
A sound discussion starts with explicit classification.

2 **Choosing the AI type is a skill**

2 diagnostic questions + a 2×2 quadrant.
A tool you'll apply in the seminars.

3 **Goal-setting stays with the human**

Every class of error needs a human loop.
The «AI / not-AI» boundary is your engineering zone.

The core question of the lecture: where does AI work, where does it not, and how do you tell?

Semester map: 17 lectures, 3 modules + exam.

Module 1	Module 2	Module 3	Exam
Theory & method basics + familiar industries	*Engineering design, heavy industry, agri*	*Infrastructure, science, extraction, synthesis*	*Final assessment*
1.1 Introduction	2.1 CAD/CAM	3.1 Logistics	Exam
1.2 AI architecture	2.2 Aerospace / defense	3.2 Telecom / cybersec	
1.3 Agents, RAG, API	2.3 Manufacturing	3.3 Science	
1.4 Software development	2.4 Digital twins	3.4 Oil & gas	
1.5 Finance / retail	2.5 Agriculture ◆M2 test	3.5 Medicine	
1.6 Creative ◆M1 test		3.6 Synthesis ◆M3 test	

◆ — *module tests M1/M2/M3, at the end of each of the first three modules.*

Semester grade

$$\mathbf{100} = \mathbf{10}\ \textit{(attendance)} + \mathbf{30}\ \textit{(exam)} + \mathbf{3\times 20}\ \textit{(M1/M2/M3 tests)}$$

Module tests — at the end of each of the first three modules.

Lecture 2: «How modern large models work».

Tokens

the units the model chops text into

Embeddings

(vector representations) — addresses in a meaning space

Attention

— which parts of the input to look at

Temperature

the randomness in picking the next token

These 4 concepts explain the behavior of every modern LLM — from ChatGPT to DeepSeek.

Questions?

Thank you